A Brain-Computer Interface Control System Design Based on Deep Learning

Third Year Individual Project – Final Report

April 2026

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Contents

Abstract

Motor imagery (MI) Brain-Computer Interfaces (BCIs) remain difficult to deploy for real-world robotic control because existing approaches treat EEG decoding as a static, open-loop classification problem and depend on high-density electrode arrays. This project presents a closed-loop BCI control pipeline that addresses these limitations through three contributions: (1) a hybrid CNN and Transformer classifier (EEGTransformer) that captures both the spatial specificity and long-range temporal structure of MI signals; (2) a Deep Q-Network (DQN) reinforcement learning agent that compensates for transient classification errors by aggregating classifier evidence over time; and (3) a systematic channel reduction study that maps the 64-channel laboratory setup down to an 8-channel configuration compatible with consumer-grade OpenBCI hardware. Furthermore, an evaluation of Independent Component Analysis (ICA) showed limited benefit for motor imagery preprocessing, leading to the adoption of a bandpass-only pipeline.

Evaluated across BCI Competition IV-2a (22-channel, 4-class), IV-2b (3-channel, 2-class), and PhysioNet EEGMMIDB (64-channel, 109 subjects), the EEGTransformer achieved mean accuracies of 73.80%, 82.87%, and 88.78% (with subject-specific fine-tuning), respectively. The channel reduction study identified C3 as the most critical electrode and showed that an 8-channel subset retains 72.54% accuracy after fine-tuning. An end-to-end simulation demonstrated that the DQN agent achieves a 99% target reach rate even when driven by imperfect EEG classifications (82% accuracy), confirming that closed-loop RL effectively compensates for decoder errors. EEG classification was evaluated offline using pre-recorded datasets, while control evaluation was conducted in simulation, as this undergraduate project does not have University of Manchester ethical approval for live human subject testing.

Declaration of Originality

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Acknowledgements

The author thanks Alex Casson for feedback on the report draft and for weekly technical discussions throughout the project. Public benchmark datasets, open-source scientific software, and the University computing environment were essential to the completion of this work.

1 Introduction

1.1 Background and Motivation

Electroencephalography (EEG) is a non-invasive neuroimaging technique that records the electrical activity of the brain through electrodes placed on the scalp. When large populations of cortical neurons fire synchronously, they generate voltage fluctuations on the order of microvolts (μV), which surface electrodes can detect with millisecond temporal resolution. EEG is widely used in clinical neuroscience for diagnosing epilepsy, sleep disorders, and other neurological conditions, and has become the predominant recording modality for Brain-Computer Interfaces (BCIs) because of its portability, low cost, and high temporal resolution relative to other neuroimaging techniques such as functional magnetic resonance imaging (fMRI) or magnetoencephalography (MEG) [1].

A Brain-Computer Interface (BCI) is a system that translates neural signals directly into commands for external devices, bypassing the body's normal neuromuscular output pathways. BCIs offer a communication and control channel for individuals with severe motor disabilities—such as amyotrophic lateral sclerosis (ALS), spinal cord injury, or locked-in syndrome—who cannot use conventional interfaces. Beyond assistive technology, BCIs are increasingly explored for rehabilitation, gaming, and hands-free device control [5].

Motor Imagery (MI) is a mental process in which an individual imagines performing a movement (e.g., clenching the left hand, moving the right foot) without actually executing it. During MI, the sensorimotor cortex produces characteristic changes in neural oscillations known as Event-Related Desynchronisation (ERD) and Event-Related Synchronisation (ERS). Specifically, imagining a movement of one limb causes a suppression (ERD) of the Mu rhythm (8–13 Hz) and Beta rhythm (13–30 Hz) over the contralateral motor cortex, and a simultaneous enhancement (ERS) over the ipsilateral cortex. These spatially and spectrally specific patterns provide the neural signatures that MI-BCI systems exploit for classification [1].

MI-based BCIs are particularly attractive because they require no external stimulation (unlike P300 or SSVEP paradigms) and rely entirely on the user's voluntary mental activity. However, translating MI-BCI systems into reliable, real-time robotic controllers remains an open problem driven by three fundamental challenges:

Signal non-stationarity. EEG patterns shift across sessions, days, and subjects due to variations in electrode impedance, fatigue, and cognitive strategy [1]. A decoder calibrated on one recording session therefore degrades when applied to subsequent sessions or new users, limiting the practical lifespan of static models.

Open-loop control constraints. Conventional BCI pipelines operate in open loop. Each classified epoch is mapped to a fixed command, meaning the system cannot recover from a misclassification once it occurs. In robotic manipulation, where smooth, corrective trajectories are essential, a single erroneous command can destabilise an entire motion sequence.

Hardware scalability. Most high-performing MI decoders are developed on 64-channel research-grade amplifiers, yet portable deployment requires consumer-grade devices such as the OpenBCI Cyton (8 channels). Naively discarding channels forfeits the spatial information on which these decoders depend. Therefore, channel reduction requires an empirical estimate of which electrodes carry the most discriminative signal.

1.2 Aims and Objectives

The overall aim of this project is to **design, implement, and validate a closed-loop Brain-Computer Interface control system** that translates motor imagery EEG signals into robotic arm commands using deep learning and reinforcement learning, while remaining deployable on consumer-grade hardware.

To achieve this aim, five objectives are pursued. On the classification side, the project develops a hybrid CNN-Transformer architecture (EEGTransformer) that combines convolutional spatial filtering with Transformer-based temporal modelling, evaluated across BCI Competition IV-2a, IV-2b, and PhysioNet EEGMMIDB to assess generalisation across electrode densities, class counts, and subject populations. A two-stage transfer learning strategy, comprising cross-subject pre-training followed by subject-specific fine-tuning, is implemented to bridge the cross-subject generalisation gap while minimising per-user calibration burden. A systematic channel reduction study uses ablation-based importance analysis to identify an 8-channel electrode subset compatible with the OpenBCI Cyton and quantify the trade-off between channel count and classification accuracy.

On the control side, a closed-loop Deep Q-Network (DQN) reinforcement learning agent integrates noisy MI predictions over time and learns corrective motor policies, decoupling control success from instantaneous classification accuracy. Finally, the complete end-to-end pipeline, spanning EEG acquisition, preprocessing, classification, RL control, and robotic actuation, is evaluated in simulation under imperfect EEG decoding conditions.

1.3 Proposed Approach

These limitations motivate a system that combines complementary strengths, and this project contributes in three areas.

At the classification level, an EEGTransformer architecture pairs an EEGNet-inspired convolutional front-end with a multi-layer Transformer encoder. The convolutional stage provides the spatial inductive biases that pure Transformers lack, while the Transformer captures the long-range temporal dependencies that CNNs and Common Spatial Patterns (CSP) miss. This combination retains the data-driven spatial filtering of EEGNet while adding global temporal context, addressing the single-trial isolation problem of purely convolutional models without incurring the training instability of recurrent neural networks (RNNs).

At the control level, rather than treating classification as the final output, a DQN agent consumes the classifier's predictions as part of a richer state representation and learns to produce corrective motor commands through reward-driven policy optimisation. By aggregating evidence over multiple time steps, the agent can override transient misclassifications, directly addressing the open-loop fragility of conventional pipelines.

At the deployment level, a systematic ablation study quantifies each electrode's contribution to classification, enabling evidence-based selection of an 8-channel subset compatible with the OpenBCI Cyton. Combined with subject-specific fine-tuning from a pre-trained 64-channel model, this strategy preserves practical classification performance while reducing hardware cost and setup complexity by an order of magnitude.

The system is validated across three datasets that collectively span different electrode montages (3, 22, 64 channels), class counts (2, 4), and subject pools (9–109). End-to-end control is evaluated in simulation using pre-recorded EEG data and PyBullet robotic arm simulation.

1.4 Report Structure

The remainder of the report is organised as follows. Chapter 2 reviews the main MI-EEG classification paradigms, representative benchmark datasets, and the gap addressed by this project. Chapter 3 describes the methods, including dataset handling, preprocessing, EEGTransformer design, training protocol, channel-reduction procedure, and the RL control formulation. Chapter 4 presents the experimental results and discusses their implications for classification accuracy, channel efficiency, preprocessing design, and closed-loop control. Chapter 5 summarises how the project was planned and managed. Chapter 6 concludes the report and outlines the most relevant directions for future work.

2 Literature Review

2.1 Introduction

This chapter surveys existing approaches to MI-EEG classification and the datasets commonly used for evaluation, situating the current project within the broader research landscape.

2.2 MI Classification Methods

Classical spatial filtering. The dominant classical pipeline extracts Common Spatial Patterns (CSP) features and classifies them with Linear Discriminant Analysis (LDA) [1]. CSP maximises the variance ratio between classes, producing compact spatial filters that are neurophysiologically interpretable. However, CSP assumes that the covariance structure of the data is stationary across trials, an assumption violated by session-to-session drift. As a result, CSP+LDA performance degrades rapidly without frequent recalibration. Filter Bank CSP (FBCSP) [13] extends CSP by applying multiple frequency sub-bands and selecting the most discriminative features, achieving improved robustness at the cost of higher computational complexity.

Convolutional neural networks. To relax the stationarity assumption, convolutional architectures such as EEGNet [8] and ShallowConvNet [10] learn spatial–temporal filters directly from data. EEGNet introduced depthwise and separable convolutions for compact EEG feature extraction, achieving 68–72% on standard 4-class benchmarks. ShallowConvNet applies temporal and spatial convolutions with log-variance pooling to capture band-power features. Yet these models process each trial independently: they lack any mechanism to carry information across successive epochs, making them susceptible to isolated noisy trials.

Recurrent architectures. Recurrent networks (LSTMs, GRUs) address this limitation by maintaining hidden states across time steps, enabling sequential modelling of EEG dynamics. In practice, however, their performance on long MI epochs is hampered by vanishing gradients and sensitivity to sequence length, and their strictly sequential computation prevents parallelisation during training [11].

Transformer-based models. Transformers overcome the sequential bottleneck of RNNs through self-attention, modelling arbitrary pairwise dependencies in a single forward pass.

EEG-Conformer [12] combines convolutional modules with Transformer encoders for MI classification, achieving state-of-the-art results. ATCNet [14] combines attention-based temporal convolutional networks with sliding window augmentation. EEG-TCNet [15] proposes a compact temporal convolutional architecture as an efficient alternative. Their weakness relative to CNNs is the opposite: pure self-attention lacks the spatial inductive biases that make EEGNet effective.

Reinforcement learning in BCI. A separate line of work applies RL to BCI, but primarily for offline feature selection. MARS [4] uses multi-agent RL to optimise spatial, spectral, and temporal filter banks. RLEEGNet [3] formulates EEG decoding within an MDP framework. However, neither system closes the loop with a physical actuator.

Table 1 summarises representative MI classification methods and their reported performance on the BCI Competition IV-2a benchmark.

Table 1: Comparison of MI-EEG classification methods on BCI Competition IV-2a (4-class, 22-channel). Results are mean subject-dependent accuracies (%) unless otherwise noted.

Method	Architecture Type	Accuracy (%)	Reference
CSP + LDA	Spatial filtering + linear	60–68	[1]
FBCSP + SVM	Filter bank + classifier	67.6	[13]
ShallowConvNet	Temporal–spatial CNN	68–71	[10]
EEGNet	Depthwise separable CNN	68–72	[8]
FBCNet	Filter-bank + CNN	76.2	[16]
EEG-TCNet	Temporal convolutional	77.4	[15]
EEG-Conformer	CNN + Transformer	77.7	[12]
ATCNet	Attention temporal conv.	80.5	[14]
EEGTransformer (Ours)	**CNN + Transformer**	**73.80**	**This work**

As shown in Table 1, purely convolutional approaches (EEGNet, ShallowConvNet) typically achieve 68–72%, while hybrid CNN-Transformer architectures (EEG-Conformer, ATCNet) push accuracy above 77%. The EEGTransformer proposed in this project achieves 73.80%, which is competitive with established CNN baselines and supports the use of a hybrid CNN-Transformer design. Direct comparison across studies is complicated by differences in preprocessing, data augmentation, and evaluation protocols.

2.3 Datasets Used in MI-BCI Research

The choice of evaluation dataset significantly influences the reported results and the generalisability of conclusions. Table 2 provides a comparative overview of the three datasets used in this study, together with their prevalence in the MI-BCI literature and their respective advantages and limitations.

Table 2: Comparison of MI-EEG benchmark datasets: characteristics, usage prevalence, and limitations.

Dataset	Ch.	Classes	Subjects	Advantages	Limitations
BCI Competition IV-2a [17]	22	4	9	Most widely cited MI benchmark; standardised evaluation protocol; 4-class task enables multi-class comparison	Small subject pool (9); limited demographic diversity; may overfit to benchmark
BCI Competition IV-2b [17]	3	2	9	Tests minimal-channel performance; binary task is clinically relevant	Only 3 channels; 2-class task is less challenging; same 9 subjects as 2a
PhysioNet EEGMMIDB [18]	64	4	109	Large subject pool enables cross-subject studies; high channel density; freely accessible	Inconsistent recording quality across subjects; some subjects show minimal MI responses; no standardised training/test split

BCI Competition IV-2a is by far the most widely used benchmark in the MI-BCI literature, with hundreds of published results enabling direct methodological comparison. Its 4-class paradigm (left hand, right hand, feet, tongue) provides a rigorous multi-class test, but its small participant pool (9 individuals) limits the statistical power of cross-subject analyses.

BCI Competition IV-2b uses only 3 bipolar EEG channels and a binary left/right hand task, making it valuable for evaluating algorithms under extreme channel scarcity—a scenario directly relevant to consumer-grade BCI deployment.

PhysioNet EEGMMIDB offers the largest publicly available MI dataset (109 participants, 64 channels), enabling large-scale cross-participant generalisation studies that are infeasible with 9-participant datasets. However, recording quality varies across participants, and the absence of a standardised training/test partition complicates cross-study comparison. Despite these limitations, PhysioNet's scale makes it particularly useful for training data-hungry deep learning models and for evaluating transfer learning strategies.

By evaluating across all three datasets, this project tests generalisability along three independent axes: electrode density (3 vs. 22 vs. 64 channels), classification difficulty (2 vs. 4 classes), and population diversity (9 vs. 109 subjects).

Why these datasets were selected instead of other public EEG datasets. Many public EEG resources are available, including OpenBMI / Lee2019-MI [6], the High-Gamma Dataset introduced by Schirrmeister *et al.* [10], BNCI Horizon 2020 datasets such as 001-2014 and 002-2014 [7], and the GigaScience multimodal movement dataset of Jeong *et al.* [5]. These datasets are valuable, but they were not all equally aligned with the specific research questions in this project. OpenBMI is useful for large-subject and cross-session binary left/right MI analysis, but it would mainly duplicate the 2-class role already covered by BCI IV-2b while adding a different file format and preprocessing pipeline. The High-Gamma Dataset provides dense 128-channel recordings, but its movement/rest paradigm and emphasis on high-frequency executed-movement decoding are less directly matched to this project's low-cost MI-control target. BNCI 001-2014 and 002-2014 are closely related to BCI Competition IV-2a and IV-2b, which are already used here under their established benchmark framing. The Jeong *et al.* dataset contains useful multimodal and multi-movement information, but its broader paradigm would introduce comparison factors beyond MI-EEG decoding.

The selected datasets therefore had to satisfy four practical requirements simultaneously: they had to be openly accessible, genuinely focused on motor imagery rather than a different EEG paradigm, sufficiently established in the literature to support comparison, and complementary in montage size and evaluation role. BCI IV-2a was retained because it remains the standard 4-class benchmark used for direct comparison with prior MI classifiers. BCI IV-2b was retained because its 3-channel setup provides a realistic minimal-channel stress test relevant to consumer BCI deployment. PhysioNet EEGMMIDB was retained because it is large enough to support meaningful cross-subject pre-training and transfer-learning analysis at scale. For that reason, this project prioritised a deliberately complementary three-dataset set rather than a larger but less focused collection.

2.4 From Literature to System Structure

The literature reviewed above motivates more than a standalone classifier. In most prior work, EEG preprocessing, MI decoding, and robotic control are studied as separate modules; classification papers often stop at predicted labels, while control papers frequently assume a reliable intention signal already exists. This project instead combines these strands into a closed-loop structure. Conceptually, the system has three stages: signal preparation, intention decoding, and action selection. Raw EEG must first be filtered and segmented into epochs that retain task-relevant Mu/Beta activity; a decoder must then convert those epochs into class evidence; and a controller must finally interpret that evidence in the context of the robot's current state so that occasional misclassifications do not immediately cause task failure.

This interpretation is important for reading the architecture diagram that appears at the start of Chapter 3. That figure is not introducing an unrelated block diagram; rather, it summarises how the different research strands identified in this chapter are connected in the present project. Its left-hand group corresponds to the preprocessing literature, its middle group corresponds to the hybrid CNN–Transformer decoding literature, and its right-hand group corresponds to sequential control and RL-based decision making. Presenting the system in this modular way makes clear why the project cannot be evaluated solely as an open-loop classification benchmark: the intended contribution is the integration of these stages into a single BCI-to-control pipeline.

2.5 Summary

The literature indicates that strong MI decoders must balance spatial inductive bias, temporal modelling capacity, and robustness to subject variability. It also shows that no single benchmark captures all deployment-relevant conditions. These observations motivate the hybrid EEGTransformer design and the multi-dataset evaluation protocol adopted in the next chapter.

3 Methods

3.1 Introduction

This chapter defines the experimental procedures used in the project. The emphasis is on reproducibility: datasets, preprocessing choices, model configurations, optimisation settings, evaluation splits, and the control formulation are specified here, while numerical outcomes are deferred to Chapter 4.

The system is organised into three visual groups that correspond to four functions: EEG preprocessing, deep learning classification, reinforcement learning control, and hardware integration. In Figure 1, hardware integration is shown through the BrainFlow/OpenBCI acquisition interface and the serial SO-101 control interface.

Before reading the figure, it is useful to state explicitly what each group represents. The first group takes raw EEG and converts it into filtered, fixed-length epochs suitable for learning. The second group is the EEGTransformer itself, which maps each epoch to class evidence by combining convolutional spatial filtering with Transformer-based temporal modelling. The third group is the closed-loop controller, which receives both robot-state information and classifier output, then chooses a control action rather than directly executing the raw class prediction. The diagram should therefore be read from left to right as an information-processing chain: *signal* $\rightarrow$ *decoded intention* $\rightarrow$ *control decision*. This ordering is central to the project because it explains why classifier accuracy alone is not the only relevant performance measure.

3.2 Datasets

Three datasets were used, but they were not merged into a single training corpus. Instead, each was processed and evaluated independently because the montages, class definitions, and published protocols differ. Table 3 summarises the dataset properties and the protocol used for each part of the project.

The RL control stage requires a four-command mapping (left, right, up, down). Therefore, only the 4-class datasets were used for classification-to-control integration. BCI IV-2b remained a separate

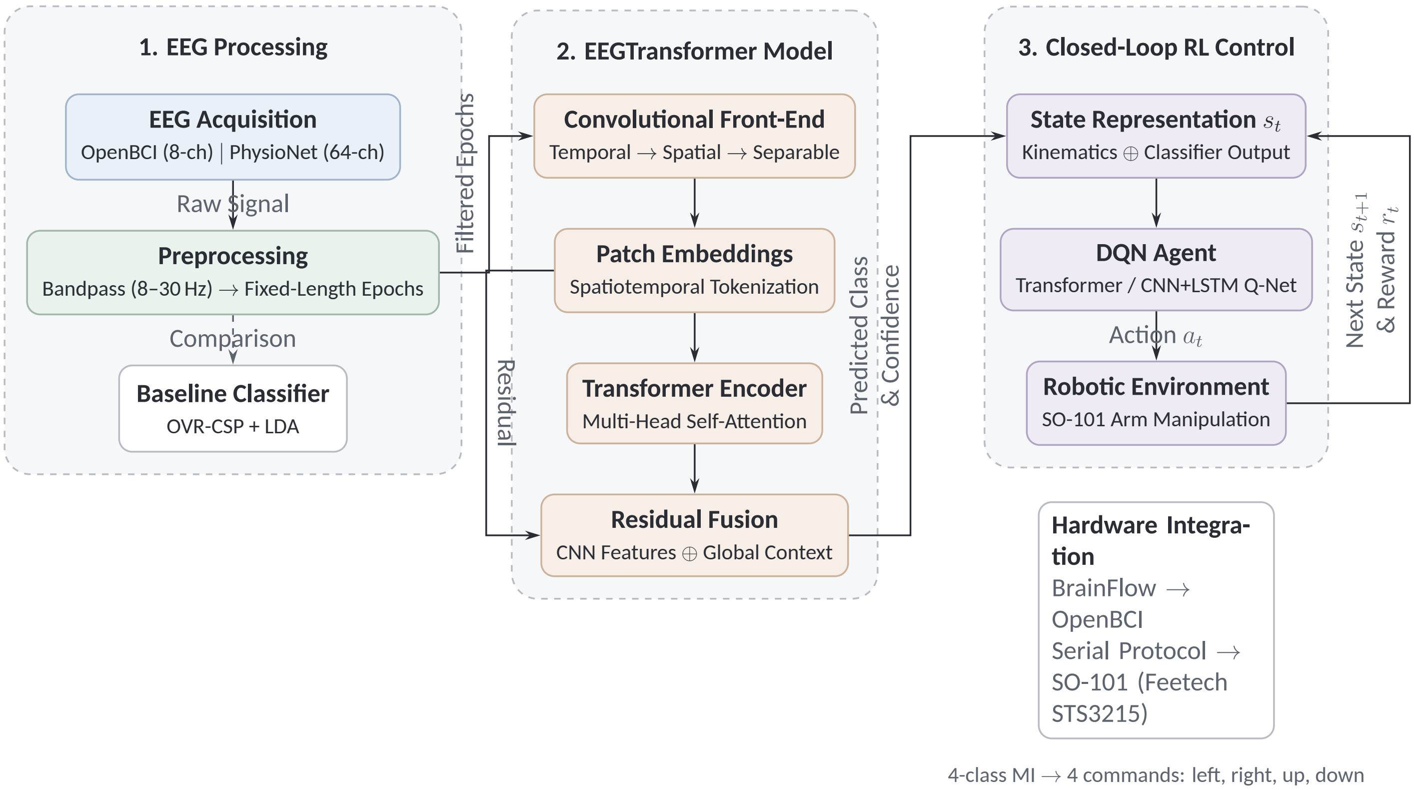

Figure 1: **System Architecture Overview.** The pipeline is shown as three visual groups spanning four functions: EEG processing, deep learning classification, reinforcement learning control, and hardware integration. Raw EEG is preprocessed, then classified by the EEGTransformer. The predicted MI class and confidence form part of the state observed by a DQN agent, whose action a_t drives the robotic environment. In the control setting used here, 4-class MI predictions are mapped to four directional commands (left, right, up, down), while hardware integration is handled through the BrainFlow/OpenBCI acquisition interface and the serial SO-101 control interface.

Table 3: Datasets used in this study and the associated evaluation protocol.

Dataset	EEG Ch.	f_s (Hz)	Classes	Participants	Protocol used in this project
BCI IV-2a	22	250	4	9	Subject-dependent evaluation using the official training/test sessions; LOSO also used for cross-participant comparison.
BCI IV-2b	3	250	2	9	Subject-dependent evaluation using the official training/test sessions; used as a minimal-channel classification benchmark only.
PhysioNet EEGMMIDB	64	160	4	109	Main pre-training and transfer-learning dataset; pooled 80/20 development split plus participant-specific fine-tuning.

2-class benchmark for testing whether the classification architecture still functions under extreme channel scarcity.

3.3 EEG Preprocessing

The preprocessing pipeline combines standard library functionality with custom project code. MNE-Python [9] was used for EDF loading, channel-name standardisation, montage assignment, and FIR filtering on PhysioNet. Custom code in this project handled label remapping, dataset-specific trial selection, resampling to a common input length, normalisation, and export into the training format used by the classifiers.

Spectral filtering. All main experiments used an 8–30 Hz bandpass, targeting the Mu and Beta rhythms associated with motor imagery. In the PhysioNet pipeline this filter was applied with MNE's FIR implementation; in the BCI IV ablation scripts the same passband was implemented with equivalent digital filtering. The filter removes slow drift and a substantial proportion of high-frequency muscle or instrumentation noise outside the task-relevant band.

Epoch definition. The common network input length was fixed at 1000 time samples. For BCI IV-2a and IV-2b, the MATLAB files already provide trial-wise segments matching the published train/test protocol. For PhysioNet, event-locked offline epochs were extracted from -0.5 s to 4.0 s around cue onset and then resampled to 1000 samples to match the CTNet input convention used throughout the project. The negative half-second was kept during offline model development because it provides a short pre-cue baseline, making it possible for the classifier to learn the transition from baseline activity into cue-aligned motor-imagery activity. The resulting 4.5 s offline window therefore preserves more experimental context than a cue-only crop while still using the same 1000-sample tensor shape as the other datasets.

Normalisation. Normalisation statistics were computed from the training partition only and then reused on held-out data. In the PhysioNet pooled script, a scalar mean and standard deviation were estimated over the complete training partition and applied unchanged to the development split. The data were therefore not re-normalised separately within each epoch. In the original CTNet-based BCI IV workflow, the official training session was further divided into training and validation subsets before optimisation.

ICA artefact removal. Independent Component Analysis (FastICA) was evaluated as an optional preprocessing stage, but it was not retained in the final operating pipeline. The decision was based on a separate ablation experiment reported in Section 4.2.6; the methodology here therefore records ICA as an evaluated variant rather than a mandatory preprocessing step.

3.4 EEG Classification: EEGTransformer

The main classifier is an EEGNet-inspired CNN–Transformer hybrid referred to in this report as *EEGTransformer*. The purpose of the convolutional front-end is to impose spatial and short-range temporal inductive bias appropriate for multi-channel EEG, while the Transformer encoder models trial-scale temporal relationships across the full motor-imagery window. Figure 2 gives the per-trial processing flow, and Table 4 summarises the main implementation settings.

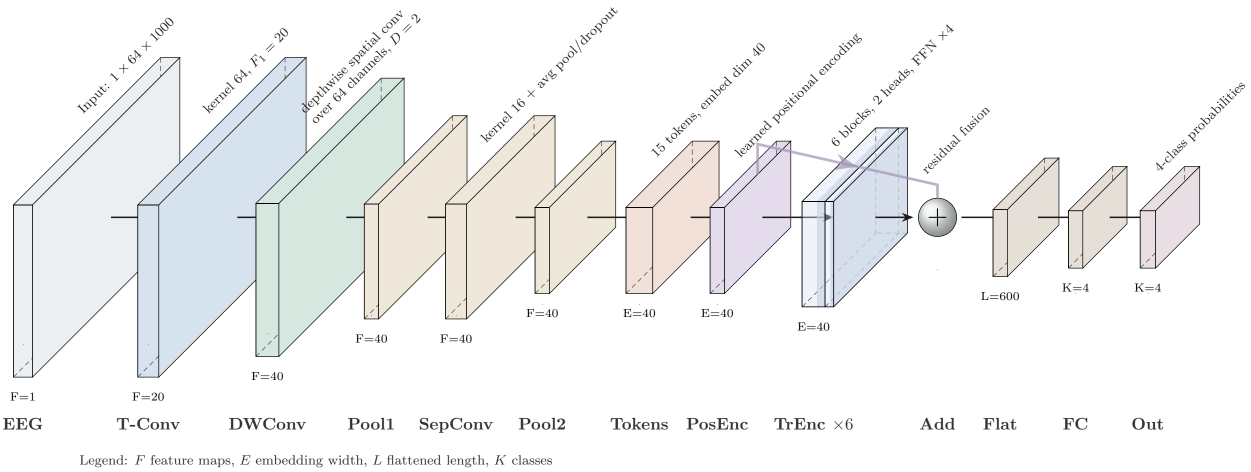

Figure 2: Per-trial EEGTransformer pipeline used for the main PhysioNet experiments. Front-edge labels denote feature-map / embedding width or classifier size, and side labels denote temporal samples or token count. The rendered architecture matches the implemented PhysioNet configuration used in code: input $1 \times 64 \times 1000$, temporal kernel 64 with $F_1 = 20$, depth multiplier $D = 2$, second temporal kernel 16, average pooling $(8, 8)$, 15 tokens of width 40, 2 attention heads across 6 encoder blocks, residual fusion, and a $600 \to 4$ linear head.

Table 4: Main EEGTransformer configuration and implementation parameters used in this project.

Component	BCI IV implementation	PhysioNet / main model
Input shape	$1 \times 22 \times 1000$ or $1 \times 3 \times 1000$	$1 \times 64 \times 1000$
Temporal convolution	kernel 64, $F_1 = 8$	kernel 64, $F_1 = 20$
Depth multiplier D	2	2
Spatial feature maps after depthwise conv.	$F_2 = 16$	$F_2 = 40$
Second temporal kernel	16	16
Average pooling	$(8, 8)$	$(8, 8)$
Token count after pooling	15	15
Transformer heads	2	2
Transformer depth	6 encoder blocks	6 encoder blocks
Token / embedding size	16	40
Positional encoding	Learnable, dropout 0.1	Learnable, dropout 0.1
Feed-forward expansion per block	$4\times$ embedding width	$4\times$ embedding width
CNN block dropout	0.25–0.50 depending on protocol	0.30
Flattened fused feature length	240 (15×16)	600 (15×40)
Classification head	Dropout 0.5 + Linear $240 \to C$	Dropout 0.5 + Linear $600 \to 4$
Residual fusion	CNN tokens + Transformer output	CNN tokens + Transformer output
Input channels	22 or 3, dataset-dependent	64 (or reduced-channel subsets in ablations)
Output classes	4 (IV-2a) or 2 (IV-2b)	4

3.4.1 Convolutional Front-End

The first stage follows an EEGNet-style factorisation [8]. A temporal convolution is applied independently across time within each channel to capture narrow-band oscillatory structure. A depthwise spatial convolution is then applied over the full electrode dimension, so each learned temporal filter produces a spatially weighted sensor combination. In practical terms, this stage plays a role analogous to data-driven spatial filtering, but without hand-engineered CSP features. A second temporal convolution and average-pooling stack reduce the sequence length while preserving the within-trial temporal ordering required by the Transformer. With a 1000-sample input and pooling sizes $(8, 8)$, the temporal axis is reduced from 1000 to 125 and then to 15 positions.

3.4.2 Transformer Encoder

The pooled convolutional output is rearranged into a sequence of tokens and augmented with positional encoding before entering a stack of Transformer encoder blocks. In the main PhysioNet configuration this produces 15 tokens of width 40, processed by 2-head self-attention repeated across 6 encoder blocks. Each block contains multi-head self-attention, a feed-forward network, residual connections, and layer normalisation. In this context, "long-range temporal dependencies" refers to relationships spanning different parts of the same offline PhysioNet epoch, extracted from -0.5 s to 4.0 s relative to cue onset (4.5 s total): pre-cue baseline activity, early cue-aligned responses, sustained ERD during imagined movement, and later within-trial recovery. These dependencies extend well beyond the local receptive field of a short convolution.

3.4.3 Residual Fusion and Classification

The Transformer output is added back to the original token sequence through a residual fusion path before classification. This preserves local information from the convolutional front-end while augmenting it with global temporal context. For the main PhysioNet model, the fused 15×40 representation is flattened to 600 features and mapped to 4 class logits by a linear classification head. The full training code, including the exact layer dimensions used in the reported experiments, is implemented in `CTNet_model.py`, `train_physionet_ctnet.py`, and the associated fine-tuning scripts.

3.4.4 Implementation Procedure

The implementation was written as an end-to-end experimental pipeline rather than as a set of isolated scripts. The model definition extends the CTNet-style CNN–Transformer structure so that the number of input channels, number of output classes, convolutional width, token width, and flattened classifier size can be selected from the dataset configuration. For the main PhysioNet setting, the configuration maps the input to 64 channels and 4 classes, uses $F_1 = 20$ with depth multiplier $D = 2$, and sets the fused token representation to 15×40 before the final linear classifier. This makes the same architectural family usable for 22-channel BCI IV-2a, 3-channel BCI IV-2b, full 64-channel PhysioNet, and reduced-channel PhysioNet variants.

The PhysioNet data pipeline was implemented by loading EDF recordings through MNE, standardising the channel montage, selecting the four motor-imagery conditions used in this project, applying the 8–30 Hz bandpass filter, resampling each epoch to 1000 samples, and storing the participant identity alongside each trial. Subjects known to be problematic in the public EEGMMIDB release were excluded before pooling. Training then applies a stratified 80/20 development split, computes normalisation statistics on the training partition only, and reuses those statistics unchanged on held-out data. This avoids leakage from the evaluation partition while retaining a single input scale for all participants.

Data augmentation was implemented using segment recombination within each class. Each 1000-sample epoch is divided into eight temporal segments; new synthetic trials are formed by stitching same-class segments sampled from different trials. This preserves the class label while exposing the model to more within-class temporal variation. During pooled training this augmentation is used with AdamW optimisation, cosine warm restarts, gradient clipping, and early stopping. During participant-specific fine-tuning, the pooled model is cloned for each participant, the convolutional front-end is frozen by default, and the Transformer/classifier layers are adapted with heavier augmentation under 5-fold stratified cross-validation. A final participant-specific model is then trained on the full calibration set after cross-validation estimates the expected accuracy.

The channel-reduction implementation follows the same pipeline but changes the channel-selection step. First, the trained 64-channel model is evaluated under leave-one-channel-out ablation to rank the effect of removing each electrode. Second, reduced-channel models are trained from concrete channel subsets rather than from abstract channel counts. This is why both ablation-ranked subsets and domain-guided motor-cortex subsets are reported: the first estimates empirical contribution, while the second keeps the resulting 8-channel layout compatible with the OpenBCI-oriented deployment target. The implementation therefore documents not only model accuracy, but also the precise data split, augmentation, normalisation, and electrode-selection decisions that produced each reported result.

3.5 Training Strategy

This subsection states the optimisation and evaluation protocol rather than the outcome. The intention is to make clear which split, validation rule, and stopping criterion corresponds to each reported experiment.

BCI IV-2a / IV-2b subject-dependent protocol. For subject-dependent experiments, the official training session (‘T‘) was used for optimisation and the official evaluation session (‘E‘) was reserved for final testing. Within the training session, 30% of trials were randomly held out as a validation subset (‘validate_ratio = 0.3‘) for model selection and early stopping. The original CTNet implementation was used with Adam optimisation, batch size 72, maximum 2000 epochs, and patience 50.

BCI IV-2a / IV-2b LOSO protocol. For cross-participant experiments on the BCI IV datasets, all trials from one held-out participant were used for testing while the remaining eight participants formed the training pool. This mirrors a standard leave-one-subject-out setting, but the validation subset was still drawn from the training pool rather than the held-out participant.

PhysioNet pooled pre-training. For the reported 109-participant PhysioNet experiment, the pooling script was run with an explicit full-subject list rather than its default '–subjects 1 2 ... 10' setting, after which a stratified 80/20 split was applied at the epoch level ('random_state = 42'). Training statistics (mean and standard deviation) were computed on the 80% partition only and then reused on the 20% hold-out partition. The main optimiser was AdamW with learning rate 5×10^{-4}, weight decay 10^{-2}, batch size 128, cosine warm restarts ($T_0 = 50, T_{mult} = 2$), gradient clipping at 1.0, and early stopping with patience 120. Segment-recombination augmentation ('InterAug') was enabled with 'n_aug = 3' and 'n_seg = 8'.

Participant-specific fine-tuning. Transfer learning was implemented by cloning the pooled PhysioNet model and fine-tuning it on one participant at a time. By default the convolutional backbone was frozen and only the remaining trainable parameters were updated. Fine-tuning used AdamW with learning rate 3×10^{-4}, weight decay 2×10^{-2}, cosine annealing, maximum 200 epochs, patience 50, and a heavier augmentation setting ('n_aug = 4', 'n_seg = 8'). Evaluation was performed with 5-fold stratified cross-validation ('shuffle = True', 'random_state = 42'), after which a final participant-specific model was trained on the full available calibration set.

Library code versus project-specific implementation. Standard libraries were used where appropriate, but the train/test protocols, transfer-learning logic, channel-selection experiments, and end-to-end integration were implemented in this project. In particular, the project code defines the data remapping, augmentation schedule, participant-wise fine-tuning workflow, and model export that are not provided by MNE-Python or PyTorch as off-the-shelf functionality.

Table 5 separates the experiments by purpose. This distinction matters because the same numerical metric does not answer every research question. Subject-dependent BCI IV experiments establish whether the architecture is competitive on standard benchmarks. PhysioNet pooled training tests whether a single model can learn from a much larger participant pool. Fine-tuning then tests whether that pooled representation can be personalised with a small calibration set. Channel reduction answers a deployment question rather than a pure classification question: how much performance remains when the electrode layout is constrained by low-cost hardware. The RL and end-to-end experiments finally shift the metric from isolated classification accuracy to task success under feedback control.

3.6 Channel Reduction Study

The channel-reduction study was designed as a protocol rather than a single table of results. Its purpose was to determine whether a 64-channel laboratory model could be reduced to an OpenBCI-compatible montage without completely redesigning the classification pipeline.

Table 5: Summary of the main experimental protocols used in the project.

Experiment	Data or environment	Optimised component	Evaluation output
BCI IV-2a classifier	Official 9-subject 22-channel benchmark	Subject-dependent EEGTransformer trained on the official training session	Per-subject 4-class accuracy on the official evaluation session
BCI IV-2b classifier	Official 9-subject 3-channel benchmark	Same CTNet family adapted to binary left/right MI	Minimal-channel 2-class accuracy and comparison with higher-density settings
PhysioNet pooled model	109-participant EEGMMIDB pool	64-channel 4-class EEGTransformer pre-training	Cross-subject held-out accuracy, precision, recall, F1, and kappa
PhysioNet fine-tuning	Ten representative participants, 90 trials each	Participant-specific adaptation from the pooled model	5-fold accuracy distribution and expected calibration benefit
Channel reduction	PhysioNet full and reduced montages	Ablation-ranked and motor-cortex channel subsets	Accuracy–channel-count trade-off and OpenBCI-oriented 8-channel choice
RL architecture comparison	Planar reaching simulation	CNN+LSTM, Transformer, and Light Transformer DQN policies	Reach rate, mean reward, parameter count, and training time
End-to-end control	Pre-recorded EEG classifications plus simulated reaching	Interaction between imperfect decoder and DQN controller	Whether closed-loop control remains successful under realistic classifier errors
BrainFlow interface check	Synthetic-board streaming and archived integration logs	Acquisition wrapper, inference call, action mapping, and position logging	Software-interface readiness without claiming live-subject validation

3.6.1 Channel Importance Analysis

First, a trained 64-channel PhysioNet model was evaluated under leave-one-channel-out ablation. Each channel was removed in turn, and the change in held-out classification accuracy was recorded. The resulting ranking was used only as a selection heuristic; it was not treated as a replacement for neurophysiological interpretation. The top-ranked 8-channel subset produced by this procedure was {C3, CP3, Fpz, Pz, O1, F8, FCz, FC2}. This list is reported explicitly because the reduction study was intended not only to compare channel counts, but also to identify which concrete electrodes were retained at each deployment-oriented operating point.

3.6.2 Progressive Channel Reduction

Second, reduced-channel models were trained and evaluated for several subset sizes. Two selection rules were compared at each size: (1) ablation-ranked subsets derived from the importance analysis and (2) domain-guided subsets centred on canonical sensorimotor electrodes. For the practical 8-channel comparison, the ablation-ranked subset was {C3, CP3, Fpz, Pz, O1, F8, FCz, FC2}, whereas the domain-guided OpenBCI-oriented subset was {C3, C4, FC3, FC4, CP3, CP4, Cz, FCz}. In the reduced-channel training script, participants were split 80/20 by participant identity rather than by epoch, so all trials from a given participant remained on only one side of the split. The reduced models were retrained with the same CTNet family, AdamW optimisation, cosine warm restarts, and InterAug augmentation. For the final 8-channel OpenBCI-compatible setting, participant-specific fine-tuning was then reused as a recovery stage.

3.7 Reinforcement Learning Control

The reinforcement-learning stage converts classification outputs into closed-loop control. Rather than mapping each classified trial directly to a motor command, the controller is formulated as a Markov Decision Process (MDP) so that the agent can correct previous mistakes over several steps.

3.7.1 MDP Formulation

The base environment is a planar 2D reaching task derived from the robotic arm end-effector kinematics. The state consists of the current end-effector position (y, z), the target position $(y^\star, z^\star)$, and the Euclidean distance to the target, giving a 5-dimensional base observation. In the EEG-augmented setting, two more values are appended: the classifier's predicted class and its confidence, giving a 7-dimensional observation. The four discrete actions are left, right, up, and down. For end-to-end experiments, the PhysioNet labels are mapped to these actions as left $\rightarrow$ left, right $\rightarrow$ right, both fists $\rightarrow$ up, and both feet $\rightarrow$ down. The reward function combines target-reaching reward, per-step penalty, distance-improvement reward, and penalties for oscillation or boundary violations.

3.7.2 Q-Network Architectures

Three Q-network families were compared:

- **CNN+LSTM baseline**: two 1D convolutions over the state sequence, followed by a single-layer LSTM and a two-layer MLP head.
- **Transformer DQN**: linear input projection from the 5-D or 7-D state to $d_{model} = 64$, sinusoidal or learnable positional encoding, 2 Transformer encoder layers, 4 attention heads, feed-forward width 256, LayerNorm, and an output head of Linear $64 \to 64 \to 4$ with GELU and dropout.
- **Light Transformer**: a compact single-attention-layer alternative with $d_{model} = 32$, 2 heads, residual self-attention, and a shallow feed-forward block.

All three architectures operate on the same state representation so that the comparison isolates the Q-network design rather than the environment. The main RL model implementation is contained in `02_Code/Reinforcement_Learning/dqn_transformer.py`, while the training and evaluation scripts define the exploration schedule, replay logic, and environment-specific reward shaping.

3.7.3 Training Protocol

The main architecture-comparison script used Double DQN updates, a replay buffer of 100 000 transitions, batch size 64, discount factor $\gamma = 0.99$, AdamW with learning rate 3×10^{-4}, cosine learning-rate annealing, and gradient clipping at 1.0. Exploration followed a linear ϵ schedule from 1.0 to 0.05 over 1500 episodes, and the target network was updated with soft updates ($\tau = 0.005$). Each episode was capped at 100 steps. This protocol was kept constant across the CNN+LSTM, Transformer, and Light Transformer experiments.

3.8 Real-Time Pipeline and Hardware Integration

The deployment workflow is separated into offline preparation, participant calibration, and online control:

Phase 1: offline preparation. A pooled EEGTransformer is trained offline on PhysioNet. This stage produces the transferable initialisation used later for participant-specific adaptation.

Phase 2: participant calibration. When a new participant uses the system, a short calibration session is collected (approximately 80 MI trials, 20 per class). The pooled model is then fine-tuned on those data and saved as a personalised classifier. If electrode placement changes substantially or the EEG distribution drifts between sessions, this calibration step can be repeated.

Phase 3: online control loop. In the intended real-time pipeline, BrainFlow serves as the acquisition interface for an OpenBCI board, a 4-second sliding window is filtered and normalised online, the personalised EEGTransformer predicts class and confidence, and the RL policy converts this observation into a control action. The chosen action is then sent to the SO-101 arm through the Feetech STS3215 serial protocol. This 4.0 s online decision window is a deployment-oriented design

choice and should be distinguished from the 4.5 s offline PhysioNet epochs (-0.5 s to 4.0 s) used during model development. In online use there is no meaningful negative-time pre-cue baseline available before the user has started the current intention, so the control loop uses the latest 4.0 s of available EEG instead. The benefit is that the online buffer maps naturally to 1000 samples at 250 Hz and gives a repeatable control update interval; the drawback is a small train–deployment mismatch and a relatively slow command rate. This is acceptable for the simulation-first validation in this report, but it motivates shorter-window and asynchronous decoding as future work.
PyBullet provides a Gymnasium-compatible simulation environment that mirrors the arm kinematics and was used for RL development and safety-preserving testing. The simplified RL comparison reported in Chapter 4 uses a 4-action planar environment, whereas the physical control code additionally supports 8-direction stepping for smoother real-arm motion during sim-to-real deployment experiments.

Online inference wrapper. The real-time control scripts separate acquisition, inference, policy selection, and actuation so that each stage can be tested independently. The BrainFlow wrapper maintains a rolling board buffer, extracts a fixed-duration EEG window, resamples it to the model's expected 1000-sample input length when necessary, applies normalisation, and reshapes the result to the CTNet tensor format. In synthetic-board tests this made it possible to exercise the complete timing and data-shape path without requiring live participants. The same wrapper also records predicted class, confidence, selected action, and position after each control step, which is why the archived integration runs can be inspected after execution rather than being treated as opaque demonstrations.

Serial-control abstraction. The SO-101 interface was implemented as a separate serial driver around the Feetech STS3215 communication protocol. At the report level, the important design decision is that the RL policy does not write raw motor packets directly. Instead, policy actions are first converted into bounded workspace updates, then into joint-level targets through the physical-control environment. This abstraction allowed the same high-level policy logic to be tested in PyBullet, synthetic BrainFlow runs, and serial-interface experiments. It also provided a natural location for practical safety controls, including velocity limiting, soft workspace limits, home/return poses, and an immediate power-off procedure during physical tests.

Why simulation remained the main validation route. The hardware interface was implemented and exercised, but the main scientific validation in this report remains offline and simulation-based. This is deliberate. The EEG classifier can be evaluated rigorously on public datasets without ethical approval, and the RL controller can be stress-tested in repeatable simulation before any live-user or physical-arm experiment. The resulting pipeline therefore demonstrates software feasibility and integration readiness while keeping human-subject validation for future work with appropriate approval.

3.9 Baseline: OVR-CSP + LDA

One-vs-Rest CSP with shrinkage LDA serves as the classical baseline. The baseline was evaluated with 5-fold stratified cross-validation on the same participant-specific data used for the corresponding deep model comparison. This provides a conventional non-deep-learning reference point against which the EEGTransformer can be interpreted.

3.10 Summary

Taken together, these methods define a pipeline that can be reproduced from raw benchmark data through to simulated robotic control. The chapter has specified how the datasets were partitioned, how the EEG was filtered and normalised, how the classifier and RL agent were configured, and how reduced-channel deployment scenarios were evaluated.

4 Results and Discussion

4.1 Introduction

This chapter presents the experimental outcomes produced by the methods of Chapter 3 and interprets them in relation to the research objectives. Subsections 4.2.1–4.2.6 report the quantitative results, while the subsequent discussion relates those findings to existing MI-BCI literature and to the deployment goals of the project.

4.2 Results

4.2.1 EEG Classification Performance

BCI Competition IV-2a (22-Channel, 4-Class). Table 6 presents per-participant classification accuracies. The EEGTransformer achieved a mean of **73.80%** $\pm$ **7.07%**, with a peak of 86.11% on Participant 3. The OVR-CSP+LDA baseline reached 74.27% on Participant 1 under 5-fold CV, and improved to 84.7% when restricted to binary left/right classification, reflecting the reduced difficulty of a 2-class task.

Table 6: Subject-dependent classification accuracy (%) on BCI IV-2a (4-class MI).

Method	S1	S2	S3	S4	S5	S6	S7	S8	S9	Mean
OVR-CSP+LDA	74.27 $\pm$ 4.52 (S1 only, 5-fold CV)									—
EEGTransformer	73.6	65.6	86.1	77.1	69.8	63.5	68.8	80.6	79.2	**73.80**

BCI Competition IV-2b (3-Channel, 2-Class). Under extreme channel scarcity, the EEGTransformer achieved a mean of **82.87%** $\pm$ **8.01%** (Table 7). The strong performance on only three electrodes confirms that the depthwise spatial convolution can extract discriminative spatial patterns even from a minimal montage, because the binary left–right task produces pronounced lateralised ERD that is detectable at C3, Cz, and C4.

Table 7: Subject-dependent classification accuracy (%) on BCI IV-2b (2-class MI).

Method	S1	S2	S3	S4	S5	S6	S7	S8	S9	Mean
EEGTransformer	69.7	68.9	86.3	83.8	84.4	83.1	89.4	95.0	85.3	**82.87**

PhysioNet EEGMMIDB (64-Channel, 4-Class). Cross-participant training on 109 participants yielded 56.54% accuracy (Table 8), reflecting the substantial inter-individual variability in cortical anatomy and MI strategy. Participant-specific fine-tuning from the pre-trained model improved this to **88.78%** (a 32-percentage-point gain), suggesting that the pooled encoder learns representations that can be adapted with relatively little individual data.

Table 8: Classification accuracy on PhysioNet EEGMMIDB (4-class MI, 109 subjects).

Training Paradigm	Accuracy (%)	Details
Cross-subject (pooled, 109 subjects)	56.54	Precision 58.1, Recall 56.5, F1 56.8, κ = 0.42
Subject-specific fine-tuning (10 subjects)	**88.78**	Range: 75.6–97.8%, 5-fold CV

Per-participant fine-tuning results (Table 9) show that the highest-performing participants (S7, S48, S43, S70) exceed 95%, while the lowest (S50, S46) still surpass 75%, indicating that the transfer-learning protocol is effective across a range of individual MI abilities.

Table 9: Per-subject fine-tuning accuracy (%) on PhysioNet (64 channels, 5-fold CV).

Subject	Mean Accuracy (%)	Std (%)
S7	97.78	2.72
S48	96.67	4.44
S43	95.56	6.48
S70	96.67	4.44
S55	88.89	11.65
S3	87.78	9.56
S9	87.78	8.16
S38	84.44	6.48
S50	75.56	4.44
S46	76.67	11.33
Average	**88.78**	—

Figure 3 visualises the participant-specific fine-tuning results. Each participant contributed 90 four-class MI trials, evaluated with 5-fold stratified cross-validation. The fold structure is important because it prevents the fine-tuning result from depending on a single favourable train/test split: each fold holds out 18 trials while training on the remaining 72. The variation between subjects is substantial. Strong participants such as S7, S48, S43, and S70 exceed 95%, whereas S50 and S46 remain below 80%. This supports the central argument that a pooled cross-subject representation is useful, but that subject-specific adaptation is still necessary because MI separability varies strongly across individuals.

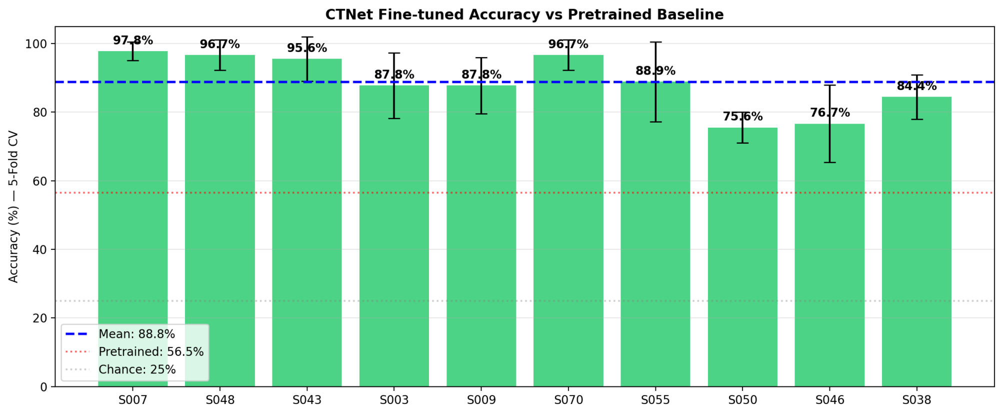

Figure 3: Participant-specific PhysioNet fine-tuning results for the 64-channel model. The plot corresponds to the 5-fold accuracies summarised in Table 9 and shows that transfer learning produces consistently high accuracy, but not uniformly across participants.

The fine-tuning workflow also helped define a realistic calibration assumption for the proposed system. A new user would not be expected to provide hundreds of trials before the system becomes useful; instead, the project assumes a short calibration session of approximately 80–90 trials. The present results suggest that this amount of participant-specific data can recover a large portion of the gap between pooled cross-subject performance and personalised performance. However, the lower-scoring participants also show why future work should investigate online adaptation rather than relying on a single fixed calibration model.

4.2.2 Channel Reduction Results

Table 10 reports cross-subject accuracy at each channel count. Performance remains comparatively stable from 64 to 16 channels, after which the ablation-ranked subset drops more noticeably. The domain-guided motor-cortex subsets show a less monotonic pattern, with the 4-channel motor-cortex configuration slightly exceeding the 8-channel motor-cortex result. Taken together, these results indicate that sensorimotor electrodes carry most of the discriminative information, while the exact reduced-channel trade-off depends on the selection rule rather than channel count alone.

Table 10: Cross-subject accuracy (%) on PhysioNet with progressive channel reduction.

Channels	Ablation-Ranked	Motor Cortex	Selection Strategy
64	51.0	—	Full set
32	51.0	47.2	Top-32 / Motor region
16	48.8	44.9	Top-16 / Motor region
8	39.0	40.7	Top-8 / OpenBCI set
4	—	43.6	Motor cortex (C3, C4, FC3, FC4)
2	—	35.8	C3, C4 only

The channel reduction study produced two complementary observations. First, classification accuracy does not decrease monotonically with channel count. For example, the 4-channel motor-cortex subset performed better than the 8-channel domain subset in the cross-subject setting. This does not imply that fewer electrodes are intrinsically better; rather, it shows that the relationship between montage size and cross-subject accuracy is affected by channel choice, subject split, and model regularisation. Second, the ablation-ranked and domain-guided subsets answer different questions. Ablation ranking asks which channels most affect the already-trained 64-channel model, whereas the domain-guided subset asks which channels are practical for an OpenBCI-oriented headset layout.

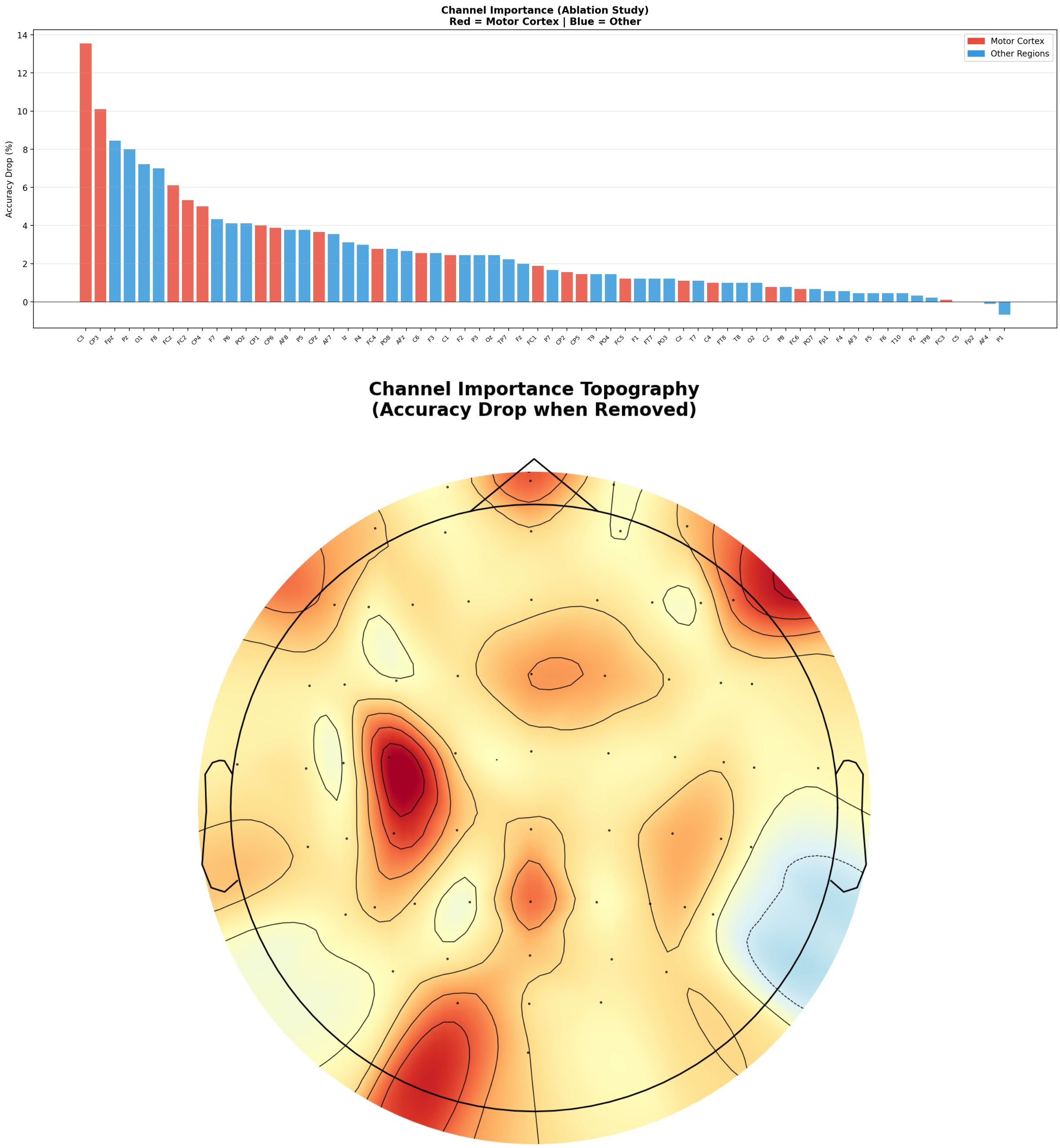

Figure 4: Ablation-based channel importance from the trained 64-channel PhysioNet model. The upper bar plot ranks the channels whose removal most affected held-out accuracy, while the lower topographic plot shows the spatial distribution of the same analysis.

Figure 4 explains why C3 appears repeatedly in the reduced-channel analysis. C3 lies over the left sensorimotor cortex and is expected to be informative for right-hand motor imagery, so its high ablation importance is neurophysiologically plausible. Other selected electrodes, such as CP3, FCz, and FC2, are also close to motor or premotor regions. At the same time, the appearance of Fpz, O1, and F8 in the purely ablation-ranked top-8 subset is a useful caution: an empirical ranking can capture dataset-specific correlations or artefact structure as well as genuine motor-cortex activity. This is why the final OpenBCI-oriented 8-channel subset was not chosen solely by the ablation list, but was constrained toward a motor-cortex montage.

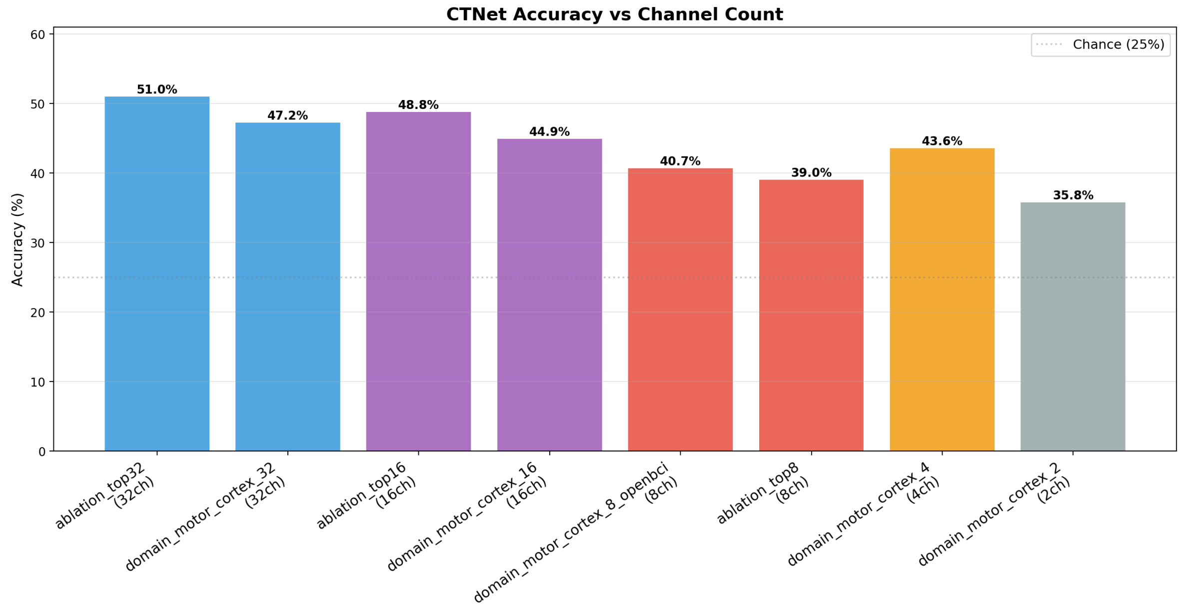

Figure 5: Cross-subject PhysioNet accuracy under progressive channel reduction. The comparison separates domain-guided motor-cortex subsets from ablation-ranked subsets, showing that channel identity matters as much as channel count.

The 8-channel motor cortex configuration (C3, C4, FC3, FC4, CP3, CP4, Cz, FCz), designed as an OpenBCI Cyton-compatible electrode set for offline evaluation, achieved 40.7% in cross-subject mode. Applying subject-specific fine-tuning recovered much of the lost performance, reaching **72.54%** on average (Table 11). This confirms that transfer learning can partially compensate for reduced spatial coverage by adapting the model to individual neural patterns. In the current online BrainFlow pipeline, however, low-channel hardware streams are still adapted to the legacy CTNet input shape before inference, so this result should be interpreted as evidence that an 8-channel montage is viable offline rather than as proof of a native end-to-end 8-channel deployment. Figure 6 shows the corresponding 8-channel fine-tuning behaviour. S7 and S48 remain close to their 64-channel performance, while S70 and S50 degrade more severely. This participant dependence suggests that channel reduction should eventually be personalised as well: a fixed OpenBCI montage is attractive for setup simplicity, but some users may require a slightly different electrode subset or a longer calibration session to reach stable control performance.

4.2.3 RL Control Performance

Table 12 compares the three DQN architectures on the simulated 2D reaching task.

Table 11: 8-channel fine-tuning accuracy (%) on PhysioNet (OpenBCI Cyton compatible).

Subject	Accuracy (%)
S7	94.44
S48	96.67
S3	67.78
S9	68.89
S70	55.56
S50	50.00
S38	74.44
Average	**72.54**

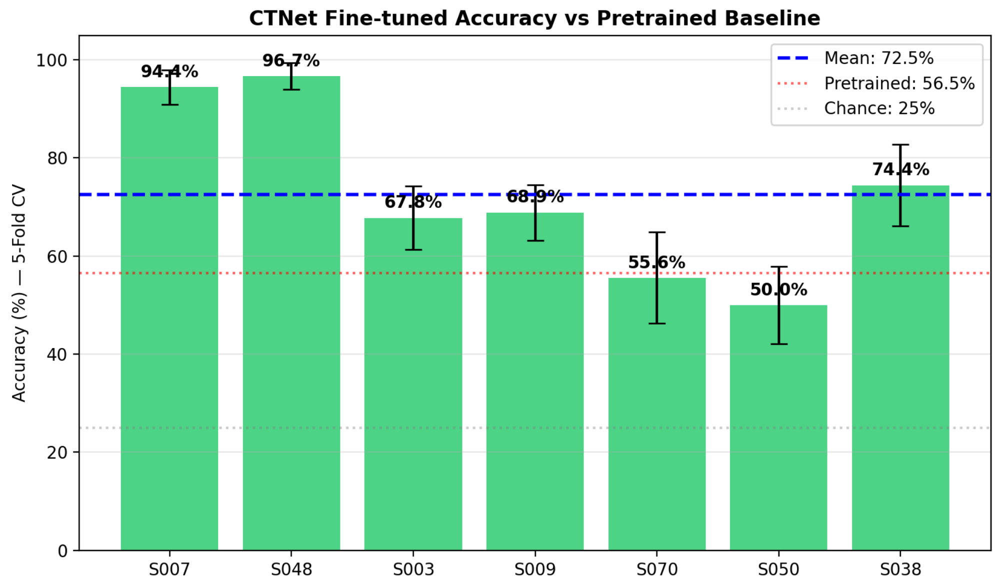

Figure 6: Participant-specific fine-tuning accuracy for the OpenBCI-compatible 8-channel model. The mean accuracy is 72.54%, but the spread across participants shows that reduced-channel deployment is strongly user-dependent.

Table 12: DQN architecture comparison on simulated 2D reaching task.

Architecture	Parameters	Training Time (s)	Final Reach Rate	Final Reward
CNN+LSTM	129,732	111.7	97%	8.79
Light Transformer	50,628	134.0	99%	10.07
Transformer DQN	**104,900**	**177.8**	**100%**	**10.39**

The Transformer DQN achieved a perfect 100% reach rate with the highest cumulative reward (10.39), outperforming both the CNN+LSTM (97%, 8.79) and Light Transformer (99%, 10.07). The CNN+LSTM reached 100% during training but dropped to 97% at evaluation, suggesting overfitting to training trajectories. The Light Transformer offers the best efficiency trade-off, achieving near-perfect control with less than half the parameters of the full Transformer.

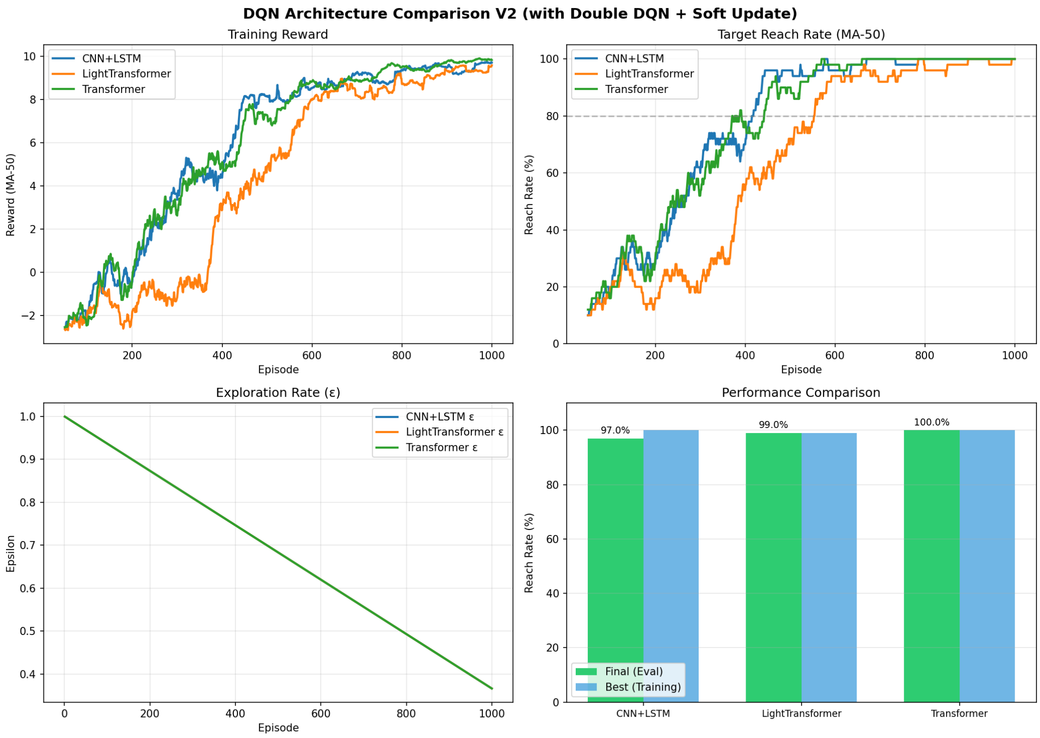

Figure 7: RL architecture comparison on the 2D reaching task. The figure complements Table 12 by showing the trade-off between reach rate, reward, parameter count, and training time for the CNN+LSTM, Light Transformer, and Transformer DQN agents.

The RL comparison was not intended to prove that the largest model is always preferable. Instead, it tested whether sequential models improve target-reaching behaviour under the same environment and reward definition. The Transformer DQN reached 100% final reach rate and the highest final reward, but it required longer training time than the CNN+LSTM and Light Transformer alternatives. The Light Transformer is therefore a useful deployment candidate: it has fewer parameters than the other models and reached 99%, only one percentage point below the full Transformer. This trade-off matters because an embedded or laptop-based BCI controller may value predictable latency and simpler inference over a marginal gain in reward.

Earlier development experiments in the source CTNet project also tested 8-direction and adaptive-step variants. These were useful for understanding smooth robotic movement, but the final report uses the 4-direction setting as the main quantitative benchmark because it aligns directly with the four MI classes used by PhysioNet: left, right, both fists/up, and both feet/down. The 8-direction experiments therefore informed the future-work direction toward richer manipulation, while the reported 4-direction experiments keep the classifier-to-control mapping transparent and reproducible.

4.2.4 End-to-End Classification-to-Control Evaluation

To validate the central claim that RL can compensate for imperfect classification, the trained EEGTransformer was used to classify 630 PhysioNet MI trials from 7 subjects, and the resulting predictions were fed into the DQN control loop. Two Transformer DQN agents were trained and evaluated: one receiving only the arm state (5-dimensional), and one augmented with the EEGTransformer's predicted class and confidence (7-dimensional). Table 13 summarises the results.

Table 13: End-to-end evaluation: RL control performance with and without EEG predictions in the agent's state. Classification accuracy of the EEGTransformer on these subjects was 82.22%.

Agent State	Dim	Reach Rate (%)	Mean Reward	Mean Steps	Train Time (s)
Arm state only	5	99.0	8.71	17.6	1130
Arm + EEG prediction	7	98.7	8.59	18.4	**668**

Both agents achieve near-perfect control ($>$98% reach rate) despite receiving classifications that are correct only 82% of the time. The EEG-augmented agent converged 41% faster during training (668 s vs. 1130 s) and reached a 100% training reach rate earlier, indicating that the classifier's directional predictions—even when occasionally wrong—provide a useful learning signal that accelerates policy acquisition. At evaluation time, however, both agents converge to comparable performance, suggesting that the Transformer DQN ultimately learns robust policies regardless of whether EEG evidence is present in the state.

Figure 8 shows why sequence-level evaluation was useful during development. A single target-reaching trial can hide oscillation, drift, or repeated wrong-direction updates. Multi-target sequences expose these behaviours because the controller must repeatedly leave one target, move toward a new target, and stabilise again. This form of testing is closer to a practical robotic-control workflow than a single static command, even though the underlying EEG evidence still comes from recorded datasets rather than live participants.

4.2.5 Software-Interface and BrainFlow Validation

In addition to dataset and simulation experiments, the project included software-interface validation for the intended online pipeline. These tests used the BrainFlow synthetic board and archived EEG-driven simulation runs, not live OpenBCI recordings. The purpose was to verify that the acquisition wrapper, fixed-window epoch extraction, classifier call, policy action, and position update could execute as a continuous loop with logged intermediate states.

The BrainFlow validation should not be over-interpreted as a live BCI experiment. It does not prove that a new participant wearing OpenBCI electrodes can control the arm in real time. Its contribution is narrower but still important: it verifies that the software interface can pass data through the same modules that a live experiment would use. In practical engineering terms, this reduces future integration risk because the next stage can focus on ethical approval, participant recruitment, electrode placement, and live signal quality rather than debugging the entire software loop from scratch.

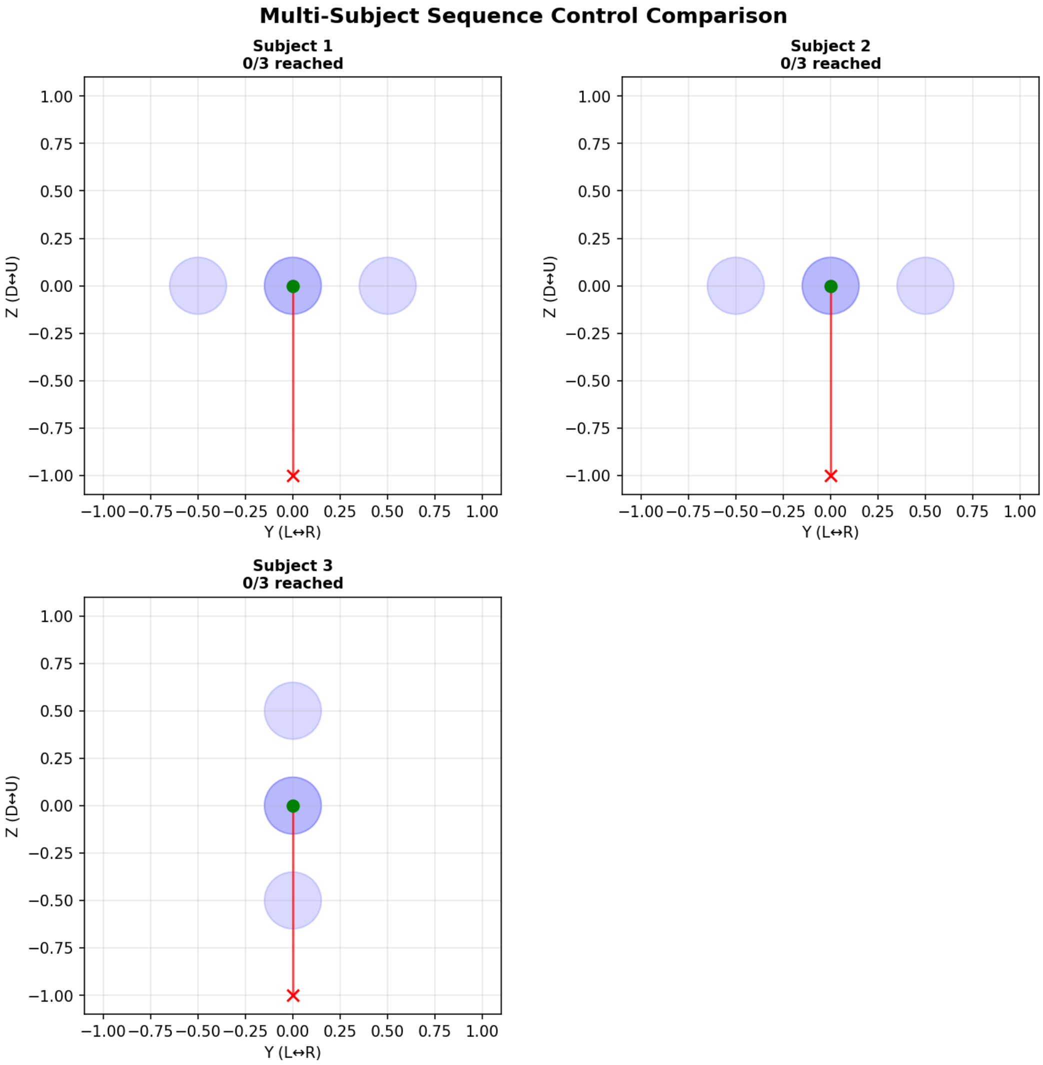

Figure 8: Archived multi-subject classification-to-control trajectories. These sequence runs test whether decoded MI commands can drive repeated target changes rather than only one isolated reach.

Table 14: BrainFlow / physical-control interface validation using synthetic-board streaming and simulated arm motion.

Validation item	Setting	Value	Interpretation
Board source	Synthetic BrainFlow board	–	Tests online data path without live subjects
Epoch duration	Sliding window	1.0 s	Faster interface test than 4.0 s design window
Movement patterns	Horizontal, vertical, diagonal, square	6	Tests multiple trajectory shapes
Targets reached	Simulated position control	24/24	No target failure in archived runs
Total control steps	All patterns	306	Confirms repeated loop execution

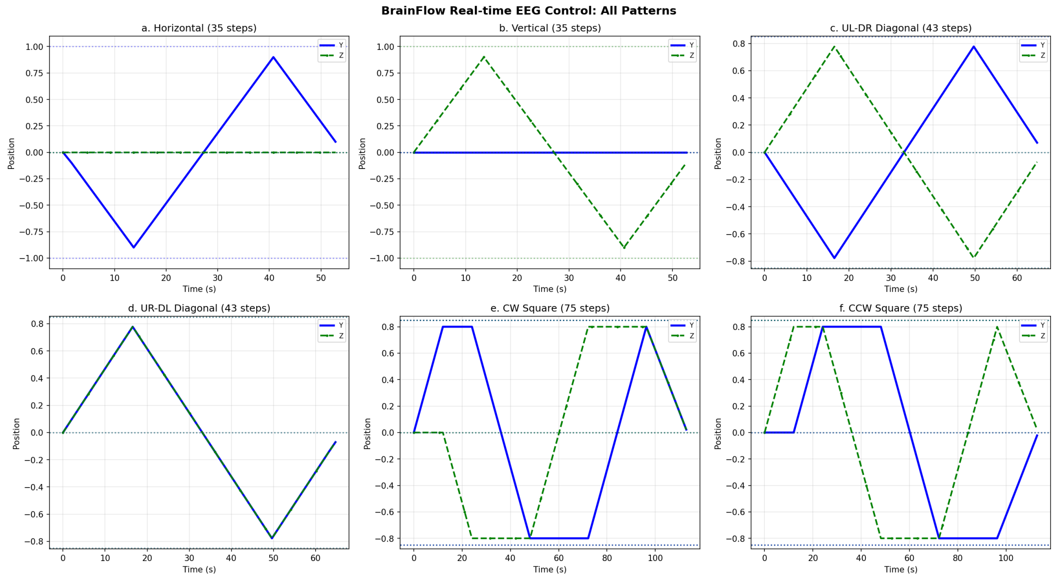

Figure 9: BrainFlow synthetic-board interface validation across six movement patterns. The figure records position over time for the archived online-control runs and verifies repeated closed-loop execution through the software interface.

4.2.6 Preprocessing Ablation: ICA and Bandpass Filtering

Table 15 reports the controlled comparison of EEGTransformer accuracy with and without ICA artefact removal on three BCI IV-2a subjects.

Table 15: Effect of ICA artefact removal on EEGTransformer classification accuracy (%).

Subject	Without ICA	With ICA	Difference
S1	76.39	73.61	-2.78
S3	79.86	81.60	$+1.74$
S7	63.54	69.10	$+5.56$
Average	**73.26**	**74.77**	$+$**1.51**

ICA improved accuracy for two of three subjects (S3: $+1.74$%; S7: $+5.56$%) but degraded it for S1 (-2.78%), yielding an average gain of only $+1.51$%. The largest improvement occurred for S7, whose lower baseline accuracy suggests greater artefact contamination in the raw data. For S1, ICA likely removed components carrying genuine motor cortex signal alongside artefacts.

Filter Ablation Study. To definitively prove the necessity of the 8–30 Hz bandpass filter, a controlled ablation experiment was conducted on PhysioNet data using the *same fine-tuning protocol* as the main results. Five subjects (S3, S7, S9, S38, S43) were fine-tuned from the 109-subject pre-trained model under two conditions: (A) with 8–30 Hz bandpass filtering, and (B) without any filtering (raw normalised data). Results are summarised in Table 16.

The filter improved accuracy for 4 out of 5 participants, with a mean advantage of $+18.44$%. For participant S43, the improvement was $+47.78$%, which is consistent with the central role of the Mu (8–12 Hz) and Beta (12–30 Hz) rhythms in motor imagery. Participant S3 showed a small decline with filtering, likely because of individual signal characteristics. Taken together, this experiment

Table 16: Filter ablation study: subject-specific fine-tuning accuracy (%) with and without 8–30 Hz bandpass filtering.

Subject	With Filter	Without Filter	Advantage
S3	55.56	62.22	-6.67
S7	74.44	74.44	$+0.00$
S9	77.78	50.00	$+27.78$
S38	54.44	31.11	$+23.33$
S43	96.67	48.89	$+47.78$
Average	**71.78**	**53.33**	$+$**18.44**

supports the use of the 8–30 Hz bandpass filter as the main preprocessing step. Combined with the marginal and inconsistent benefit of ICA ($+1.51$% average), the bandpass-only pipeline was retained as the default configuration.

4.3 Discussion

4.3.1 Why the Hybrid Architecture Outperforms Single-Paradigm Models

The EEGTransformer's competitive performance across three datasets is consistent with its two-stage design. The convolutional front-end provides spatial inductive bias that pure Transformers do not naturally have. Its depthwise spatial convolution learns data-driven spatial filters that are functionally similar to CSP and are well suited to the lateralised ERD/ERS patterns of motor imagery. The Transformer encoder, in turn, models temporal relationships across the full offline PhysioNet epoch (4.5 s from -0.5 s to 4.0 s) rather than only within local receptive fields. This interpretation is supported by the empirical results. On BCI IV-2a, the EEGTransformer's 73.80% mean exceeds the 68–72% typically reported for EEGNet and the 68–70% for FBCSP [1]. On IV-2b (3 channels), the model still achieves 82.87%, which suggests that the learned spatial filtering remains useful even in a minimal montage while the Transformer continues to provide temporal context.

4.3.2 Transfer Learning Bridges the Cross-Subject Gap

The 56.54% cross-subject accuracy on the 109-participant PhysioNet dataset reflects the well-documented challenge of EEG generalisation. Inter-individual differences in cortical folding, skull conductivity, and MI cognitive strategy create distribution shifts that no single model can fully absorb. The improvement to 88.78% after fine-tuning suggests that the cross-subject model still learns representations that transfer across participants and can then be adapted to each user's signal distribution.

Why fine-tuning helps. One plausible explanation is that the pre-trained model has already captured broad MI structure, including characteristic Mu/Beta ERD/ERS patterns, their spatial distribution over sensorimotor cortex, and the temporal profile of sustained imagery. Fine-tuning may therefore need only to adjust these shared features to participant-specific factors such as electrode placement, imagery strategy, and session-specific impedance variations. This would also help explain why a calibration set of roughly $\sim$90 trials can still produce a substantial gain.

Deployment implications. Traditional BCI calibration paradigms often train classifiers from scratch on individual data, which requires longer per-user recording. The transfer-learning approach used here may reduce that burden because the pre-trained model provides a stronger initialisation than random weights. A personalised model can then be saved and reused in later sessions, although recalibration may still be needed if electrode placement changes substantially.

4.3.3 Channel Reduction Validates Neurophysiological Expectations

The ablation study identifies C3 as the single most critical channel (13.56% accuracy drop), which is in line with established neurophysiological evidence. In the international 10-20 system, C3 overlies the left primary motor cortex hand area. During right-hand MI, contralateral ERD is often strongest at C3. More broadly, the prominence of sensorimotor channels such as C3, CP3, CP4, and FCz in the importance ranking suggests that the EEGTransformer has learned spatially meaningful representations rather than relying mainly on artefactual correlations.

The 8-channel fine-tuned accuracy of 72.54% represents a practical compromise. While lower than the 88.78% achieved with 64 channels, it remains well above the 25% chance level for 4-class classification. The end-to-end evaluation in Section 4.2.4 also shows that the DQN agent can maintain high control success when the classifier is imperfect. Taken together, these results suggest that the 8-channel electrode set remains a plausible offline foundation for closed-loop control, although real-time OpenBCI validation and a native low-channel online inference path remain future work.

4.3.4 Closing the Loop with Reinforcement Learning

The architecture comparison shows that the Transformer-based Q-networks outperform the recurrent baseline for this task (100% vs. 97% reach rate). The CNN+LSTM agent's evaluation drop from 100% during training to 97% at test time is consistent with some degree of overfitting to the training trajectories, although a more detailed ablation would be needed to isolate the cause.

The end-to-end evaluation (Section 4.2.4) tests the central hypothesis of this project: closed-loop RL can compensate for imperfect neural decoding. The EEGTransformer classified the test set at 82.22% accuracy (meaning roughly one in five commands was incorrect), yet the DQN agent still achieved a 99% target reach rate. This 17-percentage-point gap between classification accuracy and control success rate quantifies the value of closing the loop. The agent learns, through reward-driven optimisation, to correct erroneous trajectories rather than propagating them.

The comparison between EEG-augmented and state-only agents further clarifies the role of neural evidence in control. The EEG-augmented agent trained 41% faster, which suggests that even noisy directional predictions can reduce the exploration burden during learning. At convergence, however, both agents reached comparable performance. This indicates that the controller can still learn effective policies from the robotic state alone, which is encouraging for deployment under moderate classifier drift.

4.3.5 The Critical Role of Bandpass Filtering

The filter ablation study (Section 4.2.6) provides strong empirical support for using an 8–30 Hz bandpass filter in the present motor-imagery pipeline. Filtering improved accuracy by an average of $+18.44$% across 5 participants, with improvements exceeding $+20$% for 3 of them. This pattern is consistent with the neurophysiology of motor imagery, since the Mu rhythm (8–12 Hz) and Beta rhythm (12–30 Hz) lie within the retained band. Frequencies outside this range also include slow drift, mains noise, and part of the higher-frequency muscle activity that can interfere with decoding. In contrast, the ICA comparison yielded only $+1.51\%$ average improvement when ICA was applied *on top of* bandpass filtering. One plausible reason is that the 8–30 Hz filter already removes much of the ocular artefact power concentrated below 4 Hz, leaving less for ICA to improve. Another is that components within the retained Mu–Beta range may overlap with genuine sensorimotor activity, so aggressive component removal can discard useful signal as well as artefact. The participant-dependent effect of ICA (S7 improved by 5.56%, S1 degraded by 2.78%) therefore suggests that a fixed ICA policy is not always appropriate. For this reason, the bandpass-only pipeline was retained as the main preprocessing configuration.

4.3.6 Current Hardware Validation Status

At the time of writing, all core software components of the pipeline have been implemented, and their integration has been validated at dataset, simulation, or software-interface level. The EEGTransformer classifier was trained and evaluated on three public datasets (BCI IV-2a, IV-2b, PhysioNet) with consistent results; the DQN control agent was trained and evaluated in the PyBullet simulation environment, achieving a 99% reach rate; the BrainFlow acquisition interface and online control loop were exercised with synthetic-board streaming and archived EEG-driven simulation runs; and the serial communication protocol for the SO-101 arm (Feetech STS3215 servos) was implemented and verified with basic motion commands.

EEG classification, preprocessing, and channel-reduction experiments were evaluated offline using pre-recorded public datasets. Live human subject testing was not performed because this undergraduate project does not have University of Manchester ethical approval for human experimentation. The end-to-end control evaluation therefore simulates real-time use by feeding pre-recorded PhysioNet EEG data through the classification pipeline and using PyBullet for robotic arm control.

This simulation-based approach is a common step in BCI research before human-subjects deployment, and it allows the software pipeline to be tested without requiring ethical approval. The results show that the system architecture is technically feasible; future work with appropriate ethical clearance would validate these findings with live subjects.

4.3.7 Limitations

Beyond the scope of this undergraduate project, several limitations should be acknowledged. First, subject-specific fine-tuning was validated on 10 of 109 PhysioNet subjects due to computational constraints; while the results show consistent improvement across a range of individual abilities (75.6%–97.8%), validation on the full cohort would strengthen generalisability claims. Notably, this sample size is comparable to the BCI Competition IV benchmarks (9 subjects), which are the standard reference in the field. Second, fine-tuning introduces a per-user calibration step; online adaptive methods that update the model during use would eliminate this overhead. Third, the channel reduction study was conducted offline using existing datasets, so validating the 8-channel configuration with a physical OpenBCI headset and new participants remains future work requiring appropriate ethical approval. Fourth, the current online-control design uses a fixed 4-second decision window, while the offline PhysioNet training epochs span -0.5 s to 4.0 s (4.5 s total); even the 4-second online window may be too slow for time-critical applications, so shorter windows or asynchronous paradigms should be investigated.

4.4 Summary

The results show that the hybrid classifier remains competitive across heterogeneous benchmarks, that transfer learning substantially reduces the cross-subject gap, and that reduced-channel operation remains viable when paired with fine-tuning. They also show that closed-loop RL can absorb classification errors that would otherwise limit open-loop control. The next chapter summarises how the project was planned and managed before the final chapter draws together the main technical conclusions.

5 Project Management

Project management was integrated into the technical workflow rather than treated as a separate administrative task. The initial Gantt chart divided the work into dataset preparation, baseline reproduction, EEGTransformer development, transfer learning, channel reduction, RL control, and hardware integration. This sequence reduced risk by making later stages depend on validated outputs. For example, RL development began only after a stable classifier interface was available, and hardware work remained secondary to offline and simulation validation because live human testing was outside the approved project scope.

Progress was tracked through weekly supervision meetings, interim report drafts, version-controlled code, and saved experiment outputs. These checkpoints shaped the technical direction. Early 9-subject results were not sufficient for a strong cross-subject argument, so the scope was extended to the 109-participant PhysioNet evaluation. When OpenBCI deployment became a practical target, channel reduction moved from an optional extension to a core work package. The main deliverables were therefore tied to evidence: benchmark tables and saved models for classification, fine-tuning summaries for transfer learning, ablation plots for channel selection, RL logs for controller design, and BrainFlow traces for interface validation.

Risk management combined technical contingency planning with the formal University health-and-safety process. The main technical risks were excessive robotic-control complexity,

weak cross-subject generalisation, and poor reproducibility. They were mitigated by using a planar reaching task for the main RL comparison, adding participant-specific fine-tuning, and preserving scripts, model artefacts, and experiment outputs in the repository. The health-and-safety assessment covered display-screen work, electrical equipment, lone working, fire risk, software/data loss, robotic-arm electrical setup, pinch or impact hazards, object displacement, and unexpected BCI-driven arm motion. Controls included ergonomic working practice, cable and power-supply inspection, keeping liquids away from electrical equipment, maintaining a clear operating zone, reduced-speed physical tests, workspace and velocity limits, clear warning before movement, a minimum 0.5 m observer distance, and immediate power-off access. The formal artefact is stored as `05_Documentation/Risk assessment form-11314389_v3.pdf`.
The budget was kept small by using public datasets, open-source software, and existing personal computing hardware. The only project-specific purchase was an SO-101 robotic arm kit from AliExpress, costing £178.20 and including the 3D-printed arm parts, servo motors, bench vice, controller/electronics board, and communication cables. Model training used the author's existing desktop computer with an Intel i9-13900K CPU and NVIDIA RTX 4070 Ti GPU, so it is not counted as a project budget item. The supporting planning, risk, health-and-safety, CPD, and repository artefacts are listed in the appendices.

6 Conclusions and Future Work

6.1 Conclusions

This project developed a BCI-to-robot control pipeline addressing three longstanding challenges in the field. The EEGTransformer combines convolutional spatial filtering with Transformer-based temporal modelling, achieving competitive accuracies across three datasets (73.80% on IV-2a, 82.87% on IV-2b, and 88.78% on PhysioNet with fine-tuning) and generalising across electrode densities, class counts, and subject populations. Subject-specific fine-tuning bridged the 32-percentage-point cross-subject generalisation gap with minimal calibration data. The channel reduction study showed that an offline 8-channel configuration compatible with the OpenBCI Cyton retains 72.54% accuracy after fine-tuning, with C3 the most critical motor cortex electrode.
End-to-end evaluation showed that the Transformer DQN achieves 99% control success even with 82%-accurate EEG classifications, quantifying the value of closing the control loop. A controlled ICA comparison showed only marginal preprocessing benefit ($+1.51\%$), supporting the bandpass-only pipeline.
All software components, spanning EEG preprocessing, deep learning classification, reinforcement learning control, and simulated robotic arm control, have been implemented and validated. The EEG components were evaluated offline on pre-recorded datasets, and the control components in PyBullet simulation. Future work with appropriate ethical clearance should validate the closed-loop system with live human subjects and physical hardware.

6.2 Future Work

If this project were handed to a new student next year, the first recommendation would be an ethically approved live-user study. This would require an experimental protocol, participant information sheet, consent form, and risk assessment for OpenBCI data collection, followed by a small, controlled calibration dataset using the 8-channel montage identified here. That study would test whether the offline channel-reduction result transfers to real low-cost hardware.

The second recommendation is a native real-time 8-channel inference path rather than adapting the 64-channel PhysioNet workflow after the fact. The next version should train and calibrate models using the exact OpenBCI channel order, sampling rate, filtering, normalisation, and electrode-placement constraints expected during deployment. This would reduce the train–deployment mismatch and improve reproducibility.

The third recommendation is to reduce command latency. The present design uses a 4.0 s online window because it matches the 1000-sample CTNet input convention, but practical robot control would benefit from shorter overlapping windows, confidence-based rejection, and asynchronous MI detection. A useful continuation would compare 1 s, 2 s, and 4 s windows under the same classifier and RL policy to quantify the accuracy–latency trade-off.

Finally, future work should extend the controller from planar reaching toward richer SO-101 manipulation. A natural progression is to retain the present 4-direction policy as a safety baseline, then add 8-direction or 6-DOF actions after validating the live EEG decoder and emergency-stop controls. Online adaptive fine-tuning should also be investigated so that the model can track each user's current EEG distribution without requiring full recalibration.

References

[1] I. Hameed *et al.*, "Enhancing motor imagery EEG signal decoding through machine learning," *Computers in Biology and Medicine*, 2025.

[2] W. Zhao *et al.*, "A Brain-Computer Interface Control System Design Based on Deep Learning," University of Manchester, 2024.

[3] S. Nallani and G. Ramachandran, "RLEEGNet: Integrating Brain-Computer Interfaces with Adaptive AI," *arXiv:2402.09465*, 2024.

[4] D.-H. Shin *et al.*, "MARS: Multiagent reinforcement learning for spatial-spectral and temporal feature selection," *IEEE Trans. Syst. Man Cybern. Syst.*, 2024.

[5] J.-H. Jeong *et al.*, "Multimodal signal dataset for 11 intuitive movement tasks," *GigaScience*, 2020.

[6] M.-H. Lee *et al.*, "EEG dataset and OpenBMI toolbox for three BCI paradigms: an investigation into BCI illiteracy," *GigaScience*, vol. 8, no. 5, 2019.

[7] BNCI Horizon 2020, "Data sets: 001-2014 four class motor imagery and 002-2014 two class motor imagery," online database, accessed 2026.

[8] V. J. Lawhern *et al.*, "EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces," *J. Neural Eng.*, vol. 15, no. 5, 2018.

[9] A. Gramfort *et al.*, "MEG and EEG data analysis with MNE-Python," *Front. Neurosci.*, vol. 7, 2013.

[10] R. T. Schirrmeister *et al.*, "Deep learning with convolutional neural networks for EEG decoding and visualization," *Hum. Brain Mapp.*, vol. 38, no. 11, pp. 5391–5420, 2017.

[11] A. Vaswani *et al.*, "Attention is all you need," *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[12] Y. Song *et al.*, "EEG conformer: Convolutional transformer for EEG decoding and visualization," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 710–719, 2023.

[13] K. K. Ang *et al.*, "Filter bank common spatial pattern (FBCSP) in brain-computer interface," *Proc. IEEE Int. Joint Conf. Neural Networks*, pp. 2390–2397, 2008.

[14] H. Altaheri *et al.*, "Physics-informed attention temporal convolutional network for EEG-based motor imagery classification," *IEEE Trans. Ind. Inform.*, vol. 19, no. 2, pp. 2249–2258, 2022.

[15] T. M. Ingolfsson *et al.*, "EEG-TCNet: An accurate temporal convolutional network for embedded motor-imagery brain-machine interfaces," *Proc. IEEE Int. Conf. Syst. Man Cybern.*, pp. 2958–2965, 2020.

[16] R. Mane *et al.*, "FBCNet: A multi-view convolutional neural network for brain-computer interface," *arXiv:2104.01233*, 2021.

[17] M. Tangermann *et al.*, "Review of the BCI competition IV," *Front. Neurosci.*, vol. 6, p. 55, 2012.

[18] A. L. Goldberger *et al.*, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.

A Preliminary Project Proposal

Appendix A contains the preliminary project proposal submitted earlier in the academic year. The repository copy is stored as `05_Documentation/11314389_v4.pdf`. It is included in the final submission package as the original approved proposal document and is referenced here to show the initial scope against which the completed project can be compared.

B Project Plan and Gantt Chart

Appendix B contains the final updated project plan and Gantt chart used at the end of the project. The repository copy is stored as `05_Documentation/gantt_chart_v3.pdf`. This artefact provides supporting evidence for the planning decisions summarised in Chapter 5. Dated weekly progress material is also available in the repository, for example `01_Reports/week_2026_03_24.tex` and the corresponding PDF export.

C Risk Register and Health & Safety Artefact

Appendix C contains the maintained risk register together with the formal health-and-safety documentation for the project. The repository copies of the risk and health-and-safety assessment are stored as `05_Documentation/Risk assessment form-11314389_v3.docx` and `05_Documentation/Risk assessment form-11314389_v3.pdf`. These materials support the risk-management discussion in Chapter 5.

D Continuing Professional Development (CPD) Log

Appendix D contains the CPD log maintained during the project. It records the self-directed learning activities that supported the implementation work, including study in MNE-Python, PyTorch, Transformer models, reinforcement learning, and BrainFlow integration. The repository also contains dated progress material that complements the CPD record, including `01_Reports/week_2026_03_24.tex`.

E Code Repository and README

The software developed in this project is maintained in the version-controlled GitHub repository `https://github.com/SolanumLycopersicumX/3YP.git`. The repository README is available at `https://github.com/SolanumLycopersicumX/3YP/blob/main/README.md`. The repository contains the project code, experiment outputs, and supporting documentation needed to inspect the software contribution. The main authored software is organised under `02_Code/`, experimental outputs are stored under `03_Experiments/`, and supporting documents are stored under

`05_Documentation/`. Third-party or reference material is kept separate from the main authored codebase.